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A Game-Theoretic Model of Strategic Interaction in Operating Systems

Project Report

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Abstract

Operating systems traditionally assume that processes behave passively during resource allocation. However, in modern computational environments, processes may act strategically to maximize their access to limited resources such as CPU time.

This project presents a game-theoretic framework for modeling strategic interaction among processes in CPU scheduling. Each process is treated as a rational agent that selects strategies to optimize its utility, defined in terms of CPU allocation and waiting time. The study analyzes system behavior using Nash equilibrium and evaluates inefficiency through the concept of Price of Anarchy.

An incentive-compatible scheduling mechanism is proposed to improve fairness and system efficiency. The theoretical model is supported by simulation-based validation.

1 Introduction

Operating systems constitute the foundational layer of modern computing systems, orchestrating the allocation of computational resources among competing processes. Among their various responsibilities, CPU scheduling remains one of the most critical and extensively studied problems, as it directly influences system performance, responsiveness, and fairness. Classical scheduling algorithms such as First-Come First-Served (FCFS), Round Robin, and Priority Scheduling have been designed with the objective of optimizing performance metrics including waiting time, turnaround time, and throughput.

However, these traditional approaches are inherently based on the assumption that processes behave passively and adhere strictly to the scheduling policies imposed by the operating system. In contrast, contemporary computational environments—particularly distributed systems, cloud infrastructures, and multi-user platforms—introduce scenarios where processes or agents may exhibit strategic behavior in an attempt to maximize their individual benefits. Such behavior may manifest in the form of priority manipulation, resource over-requesting, or adaptive execution patterns, ultimately leading to inefficiencies, unfairness, and degradation in overall system performance.

To address this limitation, it becomes essential to incorporate a framework capable of modeling and analyzing interactions among self-interested entities. Game theory, a mathematical discipline concerned with strategic decision-making among rational agents, provides a natural and rigorous foundation for such analysis. By conceptualizing processes as players in a strategic game, where each player selects actions to maximize its utility, it becomes possible to capture the dynamics of competition and cooperation within CPU scheduling environments.

In this context, the notion of Nash equilibrium plays a central role in characterizing stable system states in which no process can unilaterally improve its utility by deviating from its chosen strategy. While such equilibria represent stable outcomes, they are not necessarily efficient from a global perspective. This discrepancy between individual rationality and collective optimality is quantitatively captured by the concept of Price of Anarchy, which measures the efficiency loss arising from decentralized, selfish behavior.

Building upon these theoretical foundations, this project proposes a comprehensive game-theoretic model of CPU scheduling, wherein processes are treated as rational agents endowed with strategic decision-making capabilities. The model aims to systematically analyze the impact of strategic behavior on system performance and to identify conditions under which inefficiencies arise.

Furthermore, the project explores the design of incentive-compatible scheduling mechanisms that align individual incentives with system-wide objectives. By introducing appropriate penalty and reward structures, such mechanisms seek to encourage cooperative behavior and mitigate the adverse effects of selfish strategies.

The proposed framework integrates formal mathematical modeling with simulation-based validation, thereby providing both analytical insights and empirical evidence. Through this approach, the project contributes to a deeper understanding of how strategic interactions

influence operating system behavior and how principled design techniques can enhance efficiency, fairness, and robustness in resource allocation.

2 Scope of the Project

This project focuses on CPU scheduling as a strategic interaction problem using game theory.

- Study classical scheduling algorithms (FCFS, Round Robin, Priority,Shortest Remaining Time(Premptive SJF))
- Model processes as strategic agents
- Analyze Nash equilibrium
- Compute Price of Anarchy
- Design incentive-compatible scheduling
- Validate results using simulation
- VCG function is used for penalizing

This work is limited to CPU scheduling and does not include memory or file systems.

3 System Model

We consider a system with n processes:

$$P = \{P_1, P_2, ..., P_n\}$$

Each process selects a strategy:

$$a_i \in \{Low, Medium, High\}$$

The CPU allocation depends on strategies:

$$CPU_i = f(a_1, a_2, ..., a_n)$$

The utility function is defined as:

$$U_i = \alpha \cdot CPU_i - \beta \cdot Waiting_i$$

where:

- CPU_i = CPU time allocated
- $Waiting_i$ = waiting time
- α, β = weighting factors

4 Objectives

The main objectives of this project are as follows:

- To model CPU scheduling as a strategic interaction problem using game theory
- To analyze the behavior of processes under selfish and cooperative strategies
- To study system stability using Nash equilibrium
- To evaluate inefficiency using the concept of Price of Anarchy
- To design an incentive-compatible scheduling mechanism
- To compare the proposed model with classical scheduling algorithms
- To validate theoretical results using simulation

5 Background and Related Work

CPU scheduling has been a fundamental area of study in operating systems. Traditional scheduling algorithms such as FCFS, Round Robin, and Priority Scheduling aim to optimize performance metrics like waiting time, turnaround time, and throughput.

However, these algorithms assume that processes do not behave strategically. In contrast, game theory studies interactions among rational agents who aim to maximize their utility. Concepts such as Nash equilibrium and social welfare are used to analyze system behavior.

Recent research has explored the application of game theory in distributed systems and resource allocation problems. Mechanism design, a branch of game theory, focuses on designing systems that encourage desirable behavior among participants.

This project builds on these ideas by applying game-theoretic modeling to CPU scheduling and analyzing the resulting system behavior.

6 Game-Theoretic Formulation

The scheduling problem is modeled as a strategic game:

- Players: Processes
- Strategies: Priority levels
- Payoff: Utility function

A Nash equilibrium occurs when no process can improve its utility by changing its strategy.

The efficiency loss is measured using Price of Anarchy:

$$PoA = \frac{W_{optimal}}{W_{equilibrium}}$$

where W is total social welfare.

7 Equilibrium Analysis

The strategic interaction among processes is analyzed using the concept of Nash equilibrium. A Nash equilibrium is defined as a strategy profile in which no process can improve its utility by unilaterally deviating from its chosen strategy.

Formally, a strategy profile $(a_1, a_2, ..., a_n)$ is a Nash equilibrium if:

$$U_i(a_i, a_{-i}) \geq U_i(a'_i, a_{-i})$$

for all processes i and for all alternative strategies a'_i.

In the context of CPU scheduling, Nash equilibrium represents a stable allocation of strategies where no process has an incentive to change its aggressiveness level. However, such equilibria are not necessarily socially optimal. Selfish behavior may lead to inefficient outcomes, where overall system performance is degraded.

This inefficiency is further analyzed using the concept of Price of Anarchy.

8 Mechanism Design

While Nash equilibrium describes stable outcomes under selfish behavior, it does not guarantee optimal system performance. To address this limitation, mechanism design is employed to construct rules that align individual incentives with global objectives.

One of the most significant mechanisms in this context is the Vickrey–Clarke–Groves (VCG) mechanism. The VCG mechanism is designed to ensure that truthful behavior is a dominant strategy for all agents.

In the proposed scheduling framework, the VCG mechanism can be interpreted as assigning a cost or payment to each process based on the externality it imposes on others. Specifically, a process that excessively consumes CPU resources reduces the utility of other processes, and this impact is quantified as a penalty.

Formally, the payment for process i can be expressed as:

$$Payment_i = Social\ Cost_{-i} - Social\ Cost$$

where:

- $Social\ Cost$ is the total cost with all processes
- $Social\ Cost_{-i}$ is the cost without process i

By incorporating such a mechanism into the scheduler, processes are incentivized to act truthfully and avoid aggressive strategies that harm overall system efficiency.

Thus, the VCG-based approach helps in achieving a more efficient and fair allocation of CPU resources, reducing the inefficiency observed in purely selfish equilibria.

9 Simulation and Implementation

The proposed game-theoretic CPU scheduling model is implemented using an event-driven simulation approach in C++. The simulator models processes as strategic agents that dynamically select strategies to maximize their utility.

Each process is assigned parameters such as arrival time, burst time, and an initial strategy. The strategies represent different levels of aggressiveness, such as Low, Medium, and High priority behavior. Based on the selected strategies, the scheduler allocates CPU time and computes performance metrics including waiting time and turnaround time.

To incorporate game-theoretic analysis, the simulation iteratively evaluates the utility of each process under different strategy profiles. At each step, a process is allowed to unilaterally change its strategy if doing so improves its utility.

The system converges to a Nash equilibrium when no process can further improve its utility by changing its strategy independently. This is implemented using a best-response dynamic, where each process repeatedly updates its strategy based on the current strategies of other processes.

The equilibrium state is recorded along with the corresponding system performance metrics. These results are then compared with the optimal (cooperative) solution to evaluate inefficiency using the Price of Anarchy.

Additionally, the simulation incorporates an incentive-based mechanism, where penalties are applied to processes that overuse CPU resources. This allows analysis of how mechanism design influences equilibrium behavior and improves fairness.

The simulation outputs are stored in structured formats such as CSV files, which are later used for visualization and performance analysis.

10 Results and Discussion

The simulation results show that strategic behavior can significantly affect system performance. Aggressive strategies may lead to reduced fairness and increased waiting time for other processes.

The proposed mechanism improves fairness and reduces inefficiency. Performance metrics such as throughput, waiting time, and fairness index are analyzed.

The results demonstrate the effectiveness of game-theoretic modeling in understanding system behavior.

11 Conclusion

This project presents a comprehensive game-theoretic framework for modeling strategic interaction in CPU scheduling within operating systems. By treating processes as rational agents with the ability to make strategic decisions, the study moves beyond traditional assumptions of passive behavior and provides a more realistic representation of modern computational environments.

The analysis demonstrates that while Nash equilibrium captures stable outcomes under selfish behavior, such equilibria may lead to inefficiencies in overall system performance. This inefficiency, quantified using the concept of Price of Anarchy, highlights the fundamental gap between individual optimization and global optimality in decentralized systems.

To address this challenge, the project incorporates principles of mechanism design, including incentive-based scheduling and VCG-inspired approaches, to align individual process behavior with system-wide objectives. The results indicate that appropriately designed mechanisms can significantly improve fairness, reduce resource monopolization, and enhance overall efficiency.

Furthermore, the integration of theoretical modeling with simulation-based validation strengthens the practical relevance of the proposed framework. The implementation of best-response dynamics and equilibrium convergence provides empirical support for the theoretical findings.

In conclusion, this work establishes that game-theoretic modeling offers a powerful and flexible approach for analyzing and improving CPU scheduling in operating systems. It opens new directions for designing intelligent, adaptive, and incentive-compatible resource allocation mechanisms in increasingly complex computing environments.